

# MACGRID™ CW 20

## HIGH TENACITY POLYESTER GEOCOMPOSITE

### MACGRID CW

20

#### Mechanical properties

|                                                |      |       |
|------------------------------------------------|------|-------|
| Tensile strength - MD<br>(EN ISO 10319)        | kN/m | 210   |
| Tolerance                                      |      | -10.0 |
| Strain at Tch strength - MD<br>(EN ISO 10319)  | %    | 10.0  |
| Tolerance                                      |      | ± 2.0 |
| Tensile strength - CMD<br>(EN ISO 10319)       | kN/m | 35    |
| Tolerance                                      |      | -5.0  |
| Strain at Tch strength - CMD<br>(EN ISO 10319) | %    | 10.0  |
| Tolerance                                      |      | ± 2.0 |

#### Physical properties

|                        |   |                                                                                                                                                                    |
|------------------------|---|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Geogrid core           |   | high tenacity polyester                                                                                                                                            |
| Polymeric coating      |   | polymeric                                                                                                                                                          |
| Roll width (standard)  | m | from 3.6 to 5.3 according to stock and production availability                                                                                                     |
| Roll length (standard) | m | 100                                                                                                                                                                |
| Durability             |   | can be exposed up to 14 days, durable in natural soil with $4 < \text{pH} < 9$ and soil temperature $< 25^{\circ}\text{C}$ for a minimum of 100 years service life |

MacGrid™ CW is a geocomposite obtained by coupling a geogrid for soil reinforcement, made from high molecular weight high tenacity polyester multifilament yarns, with a non woven geotextile of a mass that can vary from 90 g/m<sup>2</sup> to 200 g/m<sup>2</sup>, on demand.

The yarns of the geogrid are woven on tension in machine direction and finished with a polymeric coating.

MacGrid™ CW geogrids are engineered to be mechanically and chemically durable, and resistant to biological degradation.

- MD means Machine direction;
- CMD means transverse direction;
- Tch corresponds to the mean value deducted the tolerance and corresponds to the MARV at the required confidence limit;
- On reported data a typical tolerance of 3% is admitted if not differently indicated;
- Info related to tensile strength at 2%, 5% strain, and LTDS parameters are available on request;
- Special products can be manufactured on request for specific projects;
- The material is flammable;
- The material can be stocked in open space and piled until a suggested maximum height of 5 rolls.



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## MACGRID™ WG

## WOVEN HIGH TENACITY POLYESTER GEOGRID

## MACGRID WG

## 4.5S

MacGrid™ is a geogrid for soil reinforcement, made from high molecular weight, high tenacity polyester multifilament yarns.

The yarns are woven or knitted on tension in machine direction and finished with a polymeric coating.

MacGrid™ geogrids are engineered to be mechanically and chemically durable, and resistant to biological degradation.

## NOTE

1. The reported value are informative value; for certified values refer to the DoP of single items;
  2. Mean value;
  3. MD means length direction;
  4. Tch correspond to the mean value deducted the tolerance and correspond to the MARV at the required confidence limit;
  5. CMD means transverse direction;
  6. On reported data is admitted a typical tolerance of 3% if not different indicated;
  7. Different mesh size are available on request;
  8. Standard polymeric coatings are PVC, Acrylic, EVA and SBR;
  9. Variable width can be available depending on the production plan and the machines availability; please check stock position with the logistic dept;
- Info related to tensile strength at 2%, 5% and 6% strain, and LTDS parameters are available on request;
  - Special products can be manufactured on request for specific projects. The white columns are the standards products available on stock; the colored column are special products to be manufactured on request and generally not available on stock;
  - The material is flammable;
  - The material can be stocked in open space and piled until a maximum height of 5 rolls.

| Mechanical properties (typical values)         |                                                                                                                                      |                         |     |
|------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|-------------------------|-----|
| Tensile strength - MD<br>(EN ISO 10319)        | kN/m                                                                                                                                 | 50                      | 2,3 |
| Tolerance                                      |                                                                                                                                      | - 5                     |     |
| Strain at Tch strength - MD<br>(EN ISO 10319)  | %                                                                                                                                    | 10                      | 3,4 |
| Tolerance                                      |                                                                                                                                      | ± 2                     |     |
| Tensile strength - CMD<br>(EN ISO 10319)       | kN/m                                                                                                                                 | 50                      | 2,5 |
| Tolerance                                      |                                                                                                                                      | - 5                     |     |
| Strain at Tch strength - CMD<br>(EN ISO 10319) | %                                                                                                                                    | 11                      | 4,5 |
| Tolerance                                      |                                                                                                                                      | ± 2                     |     |
| Physical properties                            |                                                                                                                                      |                         |     |
| Mesh size (nominal)                            |                                                                                                                                      | from 20x20 to 35x35     | 7   |
| Geogrid core                                   |                                                                                                                                      | high tenacity polyester |     |
| Polymeric coating (standard)                   |                                                                                                                                      | polymeric               | 8   |
| Roll width (standard)                          | m                                                                                                                                    | 5.2                     | 9   |
| Roll length (standard)                         | m                                                                                                                                    | 100                     |     |
| Durability                                     | can be exposed up to 14 days, durable in natural soil with 4<pH<9 and soil temperature <25°C for a minimum of 100 years service life |                         |     |



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## MACGRID™ WG 10

## WOVEN HIGH TENACITY POLYESTER GEOGRID

## MACGRID WG

10

## Mechanical properties (typical values)

|                                                |      |        |     |
|------------------------------------------------|------|--------|-----|
| Tensile strength - MD<br>(EN ISO 10319)        | kN/m | 110    | 2,3 |
| Tolerance                                      |      | -10.00 |     |
| Strain at Tch strength - MD<br>(EN ISO 10319)  | %    | 10.00  | 3,4 |
| Tolerance                                      |      | ± 2.00 |     |
| Tensile strength - CMD<br>(EN ISO 10319)       | kN/m | 25     | 2,5 |
| Tolerance                                      |      | -5.00  |     |
| Strain at Tch strength - CMD<br>(EN ISO 10319) | %    | 10.00  | 4,5 |
| Tolerance                                      |      | ± 2.00 |     |

## Physical properties

|                              |   |                                                                                                                                      |   |
|------------------------------|---|--------------------------------------------------------------------------------------------------------------------------------------|---|
| Mesh size (nominal)          |   | from 20x20 to 35x35                                                                                                                  | 7 |
| Geogrid core                 |   | high tenacity polyester                                                                                                              |   |
| Polymeric coating (standard) |   | polymeric                                                                                                                            | 8 |
| Roll width (standard)        | m | from 3.6 to 5.3 according to stock and production availability                                                                       | 9 |
| Roll length (standard)       | m | 100                                                                                                                                  |   |
| Durability                   |   | can be exposed up to 14 days, durable in natural soil with 4<pH<9 and soil temperature <25°C for a minimum of 100 years service life |   |

MacGrid™ is a geogrid for soil reinforcement, made from high molecular weight, high tenacity polyester multifilament yarns.

The yarns are woven or knitted on tension in machine direction and finished with a polymeric coating.

MacGrid™ geogrids are engineered to be mechanically and chemically durable, and resistant to biological degradation.

## NOTE

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  2. Mean value;
  3. MD means length direction;
  4. Tch correspond to the mean value deducted the tolerance and correspond to the MARV at the required confidence limit;
  5. CMD means transverse direction;
  6. On reported data is admitted a typical tolerance of 3% if not different indicated;
  7. Different mesh size are available on request;
  8. Standard polymeric coatings are PVC, Acrylic, EVA and SBR;
  9. Variable width can be available depending on the production plan and the machines availability; please check stock position with the logistic dept;
- Info related to tensile strength at 2%, 5% and 6% strain, and LTDS parameters are available on request;
  - Special products can be manufactured on request for specific projects. The white columns are the standards products available on stock; the colored column are special products to be manufactured on request and generally not available on stock;
  - The material is flammable;
  - The material can be stocked in open space and piled until a maximum height of 5 rolls.



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# MACGRID™ WG 12

## WOVEN HIGH TENACITY POLYESTER GEOGRID

### MACGRID WG

12

#### Mechanical properties (typical values)

|                                                |      |        |     |
|------------------------------------------------|------|--------|-----|
| Tensile strength - MD<br>(EN ISO 10319)        | kN/m | 130    | 2,3 |
| Tolerance                                      |      | -10.00 |     |
| Strain at Tch strength - MD<br>(EN ISO 10319)  | %    | 10.00  | 3,4 |
| Tolerance                                      |      | ± 2.00 |     |
| Tensile strength - CMD<br>(EN ISO 10319)       | kN/m | 25     | 2,5 |
| Tolerance                                      |      | -5.00  |     |
| Strain at Tch strength - CMD<br>(EN ISO 10319) | %    | 10.00  | 4,5 |
| Tolerance                                      |      | ± 2.00 |     |

#### Physical properties

|                              |   |                                                                                                                                      |   |
|------------------------------|---|--------------------------------------------------------------------------------------------------------------------------------------|---|
| Mesh size (nominal)          |   | from 20x20 to 35x35                                                                                                                  | 7 |
| Geogrid core                 |   | high tenacity polyester                                                                                                              |   |
| Polymeric coating (standard) |   | polymeric                                                                                                                            | 8 |
| Roll width (standard)        | m | from 3.6 to 5.3 according to stock and production availability                                                                       | 9 |
| Roll length (standard)       | m | 100                                                                                                                                  |   |
| Durability                   |   | can be exposed up to 14 days, durable in natural soil with 4<pH<9 and soil temperature <25°C for a minimum of 100 years service life |   |

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#### NOTE

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  4. Tch correspond to the mean value deducted the tolerance and correspond to the MARV at the required confidence limit;
  5. CMD means transverse direction;
  6. On reported data is admitted a typical tolerance of 3% if not different indicated;
  7. Different mesh size are available on request;
  8. Standard polymeric coatings are PVC, Acrylic, EVA and SBR;
  9. Variable width can be available depending on the production plan and the machines availability; please check stock position with the logistic dept;
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## MACGRID™ WG 15

## WOVEN HIGH TENACITY POLYESTER GEOGRID

## MACGRID WG

15

## Mechanical properties (typical values)

|                                                |      |        |     |
|------------------------------------------------|------|--------|-----|
| Tensile strength - MD<br>(EN ISO 10319)        | kN/m | 160    | 2,3 |
| Tolerance                                      |      | -10.00 |     |
| Strain at Tch strength - MD<br>(EN ISO 10319)  | %    | 11.00  | 3,4 |
| Tolerance                                      |      | ± 2.00 |     |
| Tensile strength - CMD<br>(EN ISO 10319)       | kN/m | 25     | 2,5 |
| Tolerance                                      |      | -5.00  |     |
| Strain at Tch strength - CMD<br>(EN ISO 10319) | %    | 11.00  | 4,5 |
| Tolerance                                      |      | ± 2.00 |     |

## Physical properties

|                              |   |                                                                                                                                      |   |
|------------------------------|---|--------------------------------------------------------------------------------------------------------------------------------------|---|
| Mesh size (nominal)          |   | from 20x20 to 35x35                                                                                                                  | 7 |
| Geogrid core                 |   | high tenacity polyester                                                                                                              |   |
| Polymeric coating (standard) |   | polymeric                                                                                                                            | 8 |
| Roll width (standard)        | m | from 3.6 to 5.3 according to stock and production availability                                                                       | 9 |
| Roll length (standard)       | m | 100                                                                                                                                  |   |
| Durability                   |   | can be exposed up to 14 days, durable in natural soil with 4<pH<9 and soil temperature <25°C for a minimum of 100 years service life |   |

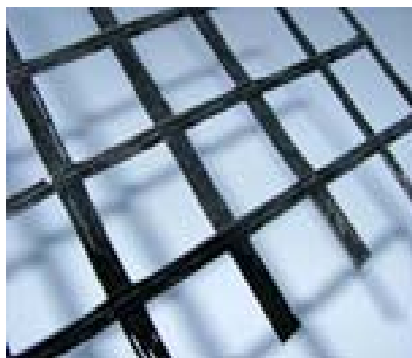
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  4. Tch correspond to the mean value deducted the tolerance and correspond to the MARV at the required confidence limit;
  5. CMD means transverse direction;
  6. On reported data is admitted a typical tolerance of 3% if not different indicated;
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# MACGRID™ WG 25

## WOVEN HIGH TENACITY POLYESTER GEOGRID

### MACGRID WG

### 25

MacGrid™ is a geogrid for soil reinforcement, made from high molecular weight, high tenacity polyester multifilament yarns.

The yarns are woven or knitted on tension in machine direction and finished with a polymeric coating.

MacGrid™ geogrids are engineered to be mechanically and chemically durable, and resistant to biological degradation.

#### NOTE

1. The reported value are informative value; for certified values refer to the DoP of single items;
  2. Mean value;
  3. MD means length direction;
  4. Tch correspond to the mean value deducted the tolerance and correspond to the MARV at the required confidence limit;
  5. CMD means transverse direction;
  6. On reported data is admitted a typical tolerance of 3% if not different indicated;
  7. Different mesh size are available on request;
  8. Standard polymeric coatings are PVC, Acrylic, EVA and SBR;
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  - The material is flammable;
  - The material can be stocked in open space and piled until a maximum height of 5 rolls.

| Mechanical properties (typical values)         |                                                                                                                                      |                                                                |     |
|------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------|-----|
| Tensile strength - MD<br>(EN ISO 10319)        | kN/m                                                                                                                                 | 260                                                            | 2,3 |
| Tolerance                                      |                                                                                                                                      | -10.00                                                         |     |
| Strain at Tch strength - MD<br>(EN ISO 10319)  | %                                                                                                                                    | 11.00                                                          | 3,4 |
| Tolerance                                      |                                                                                                                                      | ± 2.00                                                         |     |
| Tensile strength - CMD<br>(EN ISO 10319)       | kN/m                                                                                                                                 | 35                                                             | 2,5 |
| Tolerance                                      |                                                                                                                                      | -5.00                                                          |     |
| Strain at Tch strength - CMD<br>(EN ISO 10319) | %                                                                                                                                    | 11.00                                                          | 4,5 |
| Tolerance                                      |                                                                                                                                      | ± 2.00                                                         |     |
| Physical properties                            |                                                                                                                                      |                                                                |     |
| Mesh size (nominal)                            |                                                                                                                                      | from 20x20 to 35x35                                            | 7   |
| Geogrid core                                   |                                                                                                                                      | high tenacity polyester                                        |     |
| Polymeric coating (standard)                   |                                                                                                                                      | polymeric                                                      | 8   |
| Roll width (standard)                          | m                                                                                                                                    | from 3.6 to 5.3 according to stock and production availability | 9   |
| Roll length (standard)                         | m                                                                                                                                    | 100                                                            |     |
| Durability                                     | can be exposed up to 14 days, durable in natural soil with 4<pH<9 and soil temperature <25°C for a minimum of 100 years service life |                                                                |     |



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**MACGRID™ WG 50****WOVEN HIGH TENACITY POLYESTER GEOGRID****MACGRID WG****50****Mechanical properties (typical values)**

|                                                |      |        |     |
|------------------------------------------------|------|--------|-----|
| Tensile strength - MD<br>(EN ISO 10319)        | kN/m | 550    | 2,3 |
| Tolerance                                      |      | -50.00 |     |
| Strain at Tch strength - MD<br>(EN ISO 10319)  | %    | 11.00  | 3,4 |
| Tolerance                                      |      | ± 2.00 |     |
| Tensile strength - CMD<br>(EN ISO 10319)       | kN/m | 55     | 2,5 |
| Tolerance                                      |      | -5.00  |     |
| Strain at Tch strength - CMD<br>(EN ISO 10319) | %    | 11.00  | 4,5 |
| Tolerance                                      |      | ± 2.00 |     |

**Physical properties**

|                              |                                                                                                                                      |                                                                |   |
|------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------|---|
| Mesh size (nominal)          |                                                                                                                                      | from 20x20 to 35x35                                            | 7 |
| Geogrid core                 |                                                                                                                                      | high tenacity polyester                                        |   |
| Polymeric coating (standard) |                                                                                                                                      | polymeric                                                      | 8 |
| Roll width (standard)        | m                                                                                                                                    | from 3.6 to 5.3 according to stock and production availability | 9 |
| Roll length (standard)       | m                                                                                                                                    | 100                                                            |   |
| Durability                   | can be exposed up to 14 days, durable in natural soil with 4<pH<9 and soil temperature <25°C for a minimum of 100 years service life |                                                                |   |

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The yarns are woven or knitted on tension in machine direction and finished with a polymeric coating.

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